Validity Evidence for the Spiritual Harm and Abuse Scale (SHAS)

Jack Huber and Dan Koch

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Chapter 1

About this Project

This is a reproducible research project for validation of the nascent Spiritual Harm and Abuse Scale (SHAS), developed by Dan Koch. It aims to function as an evolving technical report to conceptualize and document various validity studies – and, by extension, validity evidence – for this scale. We hope some of these validity studies amount to meaningful publishable contributions to the nascent research literature in spiritual abuse and/or the literature in religion, counseling psychology, and/or psychological measurement.

1.1 Research Questions

Validation can be an ongoing agenda for research into the meanings and uses of scores from instruments. Following is an initial set of questions to guide research and scale improvement of the SHAS:

- **Item analysis and refinement** - Do some items have too many response categories?
- **Data reduction** - As written and executed, which items appear to be critical (best performing) and which appear to be redundant or noisy? What might a shorter, streamlined instrument look like?
- **Person fit** - How well does the current scale fit the population of examinees? Are there ceiling or floor effects? To what extent might the existing items under-represent the full range of abuse?
- **Scale invariance** - Do the scales mean the same thing to different groups of examinees? To what extent do these scales (abuse constructs) vary by age, gender, and/or sexuality?
- **Developmental pathway** - Beyond the current conceptualization of scales composed of redundant items, what an incremental scale of abuse with diagnostic items look like?

Chapter 2

Background of the Project

2.1 Spiritual Harm and Abuse

This is where Dan can introduce and discuss spiritual harm and abuse – his history with the topic, what it might mean in real world terms, the need for scientific study of the topic, and research to date. At a minimum he can paste in text from his extant writings and/or write new text.

2.2 The Spiritual Harm and Abuse Scale (SHAS)

This might be where Dan presents the case for a clinical measure of spiritual harm and abuse and the work he has done to date to develop the SHAS. This could be a summary of his forthcoming JSSR article.

2.3 Validity Research for the SHAS

This might be where I discuss what it means to conduct validity research on any instrument, and in particular the SHAS.

Chapter 3

Content Validity

In this chapter, we consider the content validity of the Spiritual Harm and Abuse Scale (SHAS). We begin by defining content validity and discussing its importance. Then we document the work we did to map the domain of spiritual abuse for the purpose of developing items for piloting and operational use. We conclude by considering what scope exists for more work in this area that would improve future development of SHAS instrumentation.

3.1 About Content Validity

The focus of content validity is "the relationship between the content of a test and the construct it is intended to measure" [American Educational Research Association et al., 2014]. Do they match? Does the instrument contain content that is not part of the construct? Are there parts of the content unmeasured, such that a high score still underrepresents an examinee's true level of a trait?

Content validity in this case concerns the relationship between the concept of spiritual abuse and the content of the 66 items that were piloted to form the SHAS. What is spiritual abuse? What are its important parts? What counts as evidence of it? Is there more to spiritual abuse than was captured by the 66 items?

3.2 Content Specification

"Test developers often work from a specification of the content domain. The content specification carefully describes the content in detail, often with a classification of areas of content and types of items" [American Educational Research Association et al., 2014]. For the purpose of piloting items, the domain of spiritual abuse included two constructs: external (abusive) events and internal states (of emotional harm).

3.2.1 External Events

One domain of spiritual abuse is external abuse *events*. The focus is on something that happened, or a pattern of recurring behavior on the part of someone in a place of spiritual authority that was experienced by the examinee as abusive. The first 52 items of the SHAS were drawn from this domain of possible abusive events. To each item, an examinee was offered five response categories:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of abuse. In the aggregate, scores for this collection of items could range from 52 to 260. An examinee who *never* experienced *any* of these events would score 52. At the other extreme, an examinee who scored 260 had *always* experienced *all* of these events.

3.2.2 Internal States

Another important domain of spiritual abuse is *internal states*. The focus is on the subjective experience of an examinee "as a result of negative experiences in church or Christian group settings." The questionnaire included 14 items asking examinees about their emotions in response to events. To each item, an examinee was offered the same five response categories as before:

1. Never
2. Once or twice
3. Sometimes
4. Often
5. Always

To sum each examinee's vector of responses to these items is to accumulate evidence of internal emotional *harm*. Total scores for these items could range from 14 to 70. Examinees who *never* experienced *any* of these unpleasant emotions would score 14, while examinees who scored 70 had *always* experienced *all* of the difficult emotions.

3.2.3 Additional Domains

Reise and Waller [2009] make some interesting points about the nature of the traits addressed in clinical settings which provoke thought about the trait of spiritual abuse.

3.3 Summary

What counts as spiritual abuse? What counts as evidence of it?

Chapter 4

Data Collection

The data were collected online from January 28, 2021 to February 27, 2021.

A final total of 3,222 people responded to the survey at some level. Depending on how much data are missing, this is an impressively large sample size which will provide lots of power for most simple statistics and should for the most part satisfy the large-sample requirements of the advanced statistical procedures associated with measurement work (i.e., factor analysis, IRT).

However, the data come from a convenience sample which raises some challenge of sampling bias. Dan sent the link out to [lots of people], and (if memory serves) encouraged them to send it to people *they* knew. This would seem to bias responses toward people demographically similar to Dan: white, Millennial, west coast American, middle class, highly educated, primarily Protestant and Catholic Christian, etc. This will be more apparent upon closer inspection of the demographics of the sample.

One source of examinees was people who listen to progressive Christians podcasts such as *You Have Permission* which Dan produced at the time of data collection. This is a known source of bias toward people who have some interest in Christianity along with at least some level of rejection of conservative/orthodox Christianity and/or political conservatism. However, this would be an appropriate group from which to invite participation if the target population is people who have experienced spiritual abuse.

Chapter 5

Data Preparation

This chapter documents our decisions surrounding how to prepare the data for analysis. These decisions include details on loading the necessary R packages, loading the data in R, coding or recoding of variables. Also discussed is how we deal with missing data.

5.1 Preliminary work

Load R packages. Some R users load all their packages up front. Others, perhaps for instructional reasons, load them just before calling functions, perhaps for instructional reasons. I can see merit in listing all the packages for properly citing them. Here is a code chunk for loading all the packages up front:

```
# LOAD PACKAGES
library(Hmisc)  # for variable and value labels
library(VIM)  # for missing data
library(naniar)  # for missing data
library(corrplot)  # for correlation plot
library(knitr)  # how RMarkdown compiles code into documents
```

Load data. The data are contained in an Excel file, simply named "data.xlsx", in the "data" folder of this project. The important thing is to keep all the data in the same folder within the larger "SHAS Validity" project folder.

```
# LOAD DATA
library(readxl)  # for reading the source Excel file
shas <- read_excel("data/data.xlsx")  # load Excel file
```

Select items for analysis. Which items should we include for analysis? The first 13 columns in the original data file report data from several screening items that were placed at the beginning of the questionnaire. But because they are

constant and not variable, they don't seem to have a place in scale development. Therefore it seems the important columns range from columns 14 to 97. The first set of items measure the **External Events** scale. These 52 items range from columns 14 to 35. The second set of items measure the **Internal States** scale, and these 14 items range from columns 66 to 79. The last set of items capture **demographic** variables, and these range from columns 80 to 97.

```
# SELECT DATASETS FOR ANALYSIS
all <- shas[, 14:97]  # select items, dems from columns 14-79
items <- shas[, 14:79]  # all Likert items
ext_items <- shas[, 14:35]  # External Events items
int_items <- shas[, 66:79]  # Internal States items

# REMOVE SELECTED COLUMNS
all <- subset(all, select = -c(Ext_Avg:Global_Avg))  # remove columns for now

# OTHER POSSIBLE DATASETS
write.csv(all, "data/all.csv", row.names = FALSE)  # save data for later use
write.csv(items, "data/items.csv", row.names = FALSE)
write.csv(ext_items, "data/ext_items.csv", row.names = FALSE)
write.csv(int_items, "data/int_items.csv", row.names = FALSE)
```

Labeling. This is probably the appropriate place to deal with the labeling of variables and their values. These things should be appropriately labeled to their content, wording, etc. **But I have yet to figure this part out so I will come back to it**.

```
library(Hmisc)
var.labels = c(
# VARIABLE LABELS: Items for External Events Subscale
Q7 = "My pastor/leader explicitly claiming to speak on God's behalf",
Q10 = "Being asked to give up personal, vocational, and/or educational goals by a pastor or
Q11 = "Seeing that anyone who disagreed with my church/pastor was consequently portrayed as
Q16 = "My church community abandoning me in a difficult time",
Q17 = "Being treated as "less than" because of my sexual orientation",
Q18 = "Discouragement of critical thinking by my church or pastor",
Q19 = "Behavior being excessively monitored by my pastor or group members",
Q20 = "Seeing specific members being pressured to give money despite well-known personal fi
Q21 = "Being taught that I would be risking Hell if I left my particular church/group",
Q22 = "Being expected to follow my pastor/leader's personal rules/advice around dating, mar
Q23 = "Prayer replacing needed medical interventions",
Q24 = "Being expected to consult my pastor/leader before making non-religious decisions",
Q25 = "Feeling unable to raise questions and issues",
Q26 = "Being pressured to forgive an abuser while the abuse was ongoing",
Q27 = "Being denied opportunities to serve/volunteer because of my gender",
Q28 = "Seeing the leadership or group protecting abusive individuals",
Q29 = "Feeling special when in my pastor/leader's good graces; feeling ignored when I wasn't
```

```
Q30 = "Being deterred from seeking mental health treatment, counseling and/or medication",
Q31 = "Being shunned or ignored by my pastor or church/group",
Q32 = "Feeling unable to express unhappiness",
Q34 = "Seeing Scripture used to justify physical punishment or other types of severe discipl
Q35 = "Being explicitly taught to distrust my emotions",
Q36 = "Mental or physical problems being interpreted as spiritual/moral weakness",
Q37 = "Being made to feel less spiritually advanced/mature than my pastor/leadership",
Q38 = "Having unrealistic demands placed on my moral or religious behavior",
Q39 = "Being made to feel shame over naturally occurring sexual desires (not actions)",
Q40 = "Threatening Divine punishment to keep group members in line",
Q41 = "Being denied opportunities to serve because of my sexual orientation",
Q42 = "Seeing others shamed or shunned by the pastor/leader or group",
Q43 = "Being shamed by my pastor or the group for poor spiritual/moral performance",
Q44 = "Love or acceptance only being offered if spiritual or moral performance was high",
Q45 = "Vivid descriptions of Hell, Satan, or demons being taught to young children that are
Q46 = "Being treated as "less than" because of my skin color",
Q47 = "Medical care being postponed or not given at all for religious reasons",
Q49 = "The leadership or group not only protecting but elevating abusive individuals",
Q50 = "Being treated as "less than" because of my gender",
Q51 = "Experiencing extreme pressure to take on a role of pastor, missionary, or other spiri
Q52 = "Being blamed for harm that I suffered, rather than blaming those who harmed me",
Q53 = "Seeing Scripture used to justify abusive parent-child behavior",
Q54 = "Being explicitly taught to distrust my intuitions",
Q55 = "Hearing cultural references in sermons that were not commonly known to members of my
Q56 = "Being made to feel like I was crazy or weird for having doubts or questions",
Q57 = "Developing mental or physical ailments as a result of stress from conforming to the g
Q58 = "Terror or horror being used to motivate religious decisions",
Q59 = "Seeing the pastor or group blame the victim of abuse for the abuse itself",
Q60 = "Being encouraged to remain in an abusive marriage by religious leaders",
Q61 = "Vivid descriptions of the end of the world being taught to young children that are de
Q62 = "Seeing Scripture used to justify physical violence",
Q63 = "Being shamed by my pastor or the group for raising questions or concerns",
Q64 = "Witnessing women being pressured to stay in unfaithful or abusive marriages",
Q65 = "Being "cut off" or shunned by more religious family members",
Q66 = "Seeing others treated as "less than" because of their sexual orientation",

# VARIABLE LABELS: Items for Internal States Subscale
Q68 = "Anxiety attacks triggered by religious stimuli",
Q69 = "A lack of spiritual direction or purpose",
Q70 = "Feeling betrayed by God",
Q71 = "Personally avoiding religious activities or settings to reduce distressing feelings",
Q72 = "Feeling as if God harmed me directly",
Q73 = "Self-hatred or self-loathing",
Q74 = "Sadness over the loss of my faith/religious community",
Q76 = "Feeling that I wasted years of my life in a particular church or set of beliefs",
```

```
Q77 = "Anger upon reflecting on negative religious experiences",
Q78 = "A lack of self-worth",
Q79 = "Distrust of God",
Q80 = "Feeling isolated",
Q81 = "Nightmares about my negative religious experiences",
Q82 = "Having trouble navigating life outside my church/community",

# VARIABLE LABELS for Demographics
Q84 = "Age",
Q85 = "Gender",
Q86 = "Sexual orientation",
Q87 = "Race",
Q88 = "Raised in Christian home",
Q89 = "Denomination/group of most negative experience",
Q90 = "Racial makeup of denomination/group of most negative experience",
Q91 = "Race of leadership of denomination/group of most negative experience",
Q92 = "Gender of leadership of denomination/group of most negative experience",
Q95 = "Involvement in denomination/group of most negative experience",
Q93 = "Current religious identification"
)

label(items) = as.list(var.labels[match(names(items), names(var.labels))])
```

5.2 Missing data

Missing data can be an important problem. The problem has two parts. The first is how many data are missing. Obviously the more data are missing, the larger the barrier this problem poses to analysis. The second is the mechanism of missing data. Are the missing data randomly, or is there a pattern to the missing data - like a problematic item, or a distinct group of examinees who chose not to respond? Figure 5.1 presents two plots from the VIM package for missing data [**?**].

```
# VISUALIZE AND QUANTIFY MISSING DATA
library(VIM)
a <- VIM::aggr(all, plot = FALSE)
plot(a, numbers = TRUE, prop = FALSE)
```

The first plot in Figure 5.1 is a frequency distribution of the number of missing data entries for each item. This plot reveals very few missing data for the first set of External Events items, but nearly 200 missing entries for the second set of Internal States items and demographics. The raises two questions. The first is: Who were those examinees? We know that they responded to the first 52 items, then apparently quit the survey. Why? Was it just fatigue, or is there more to know about them? The second question is what to do with the information

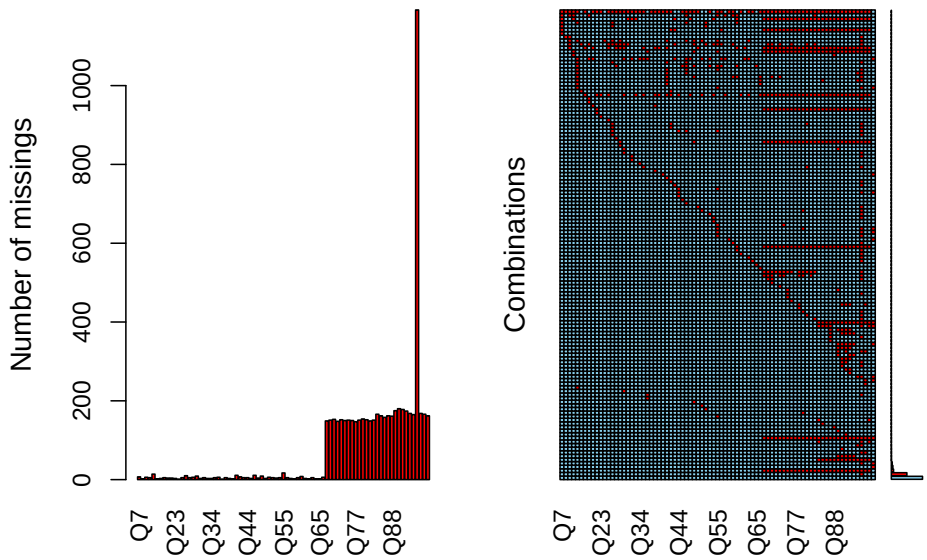


Figure 5.1: Missing Data

they actually did provide.

The second plot in Figure 5.1 illustrates the different combinations of missing entries. The blue squares represent cases with complete responses, and the red indicates missing data. The rows represent combinations of complete and missing responses. The bottom row of entirely blue squares represents complete data – examinees who responded to all of the items. The barplot on the right axis indicates that these were the most frequent combinations. The second most frequent combination was examinees who completed all of the External Events items and then quit the survey. Beyond that, there is no apparent order or pattern to the missing data.

Mechanism of missing data. McKnight et al. [2007] outline a variety of approaches for investigating the mechanisms of missing data. An appropriate first step is to test whether the data are Missing Completely At Random (MCAR) with the use of Little's test Little [1988].

```
library(naniar)
mcar_test(all)
```

```
# A tibble: 1 x 4
  statistic    df p.value missing.patterns
      <dbl> <dbl>   <dbl>            <int>
1     2089.  8663       1              117
```

This test is fairly conclusive. The p-value for the chi-square is 1, which means observed pattern of missing data do not diverge significantly from what would

be expected if the data were missing completely at random.

Were examinees triggered? As mentioned above, a possible explanation of the missing data in the Internal States items is that examinees were triggered. As after responding to the preceding 52 items calling to mind imagery of abusive external events, these examinees were **done** with the survey. A possible test of this hypothesis is an examination of the relationship between a total score of responses to the External Events items and a score of total missingness on the Internal States items. The empirical work to test this hypothesis is laid out in the following code chunk. The items measuring external events are summed into a total score such that a higher value indicates more exposure to abusive external events. The items measuring internal states are coded so that 1=missing and 0=all others response, and these items are summed into a score of total missingness.

```
# 1. MAKE A COPY OF ALL THE ITEMS JUST FOR THIS ANALYSIS
miss_set <- all

# write.csv(miss_set,'data/miss_set.csv') # save as Excel
# to explore visually

# 2. RECODE INTERNAL STATES ITEMS AS DICHOTOMOUS
# (1=MISSING, 0=ENDORSED)
miss_set$Q68[miss_set$Q68 > 0] <- 0
miss_set$Q69[miss_set$Q69 > 0] <- 0
miss_set$Q70[miss_set$Q70 > 0] <- 0
miss_set$Q71[miss_set$Q71 > 0] <- 0
miss_set$Q72[miss_set$Q72 > 0] <- 0
miss_set$Q73[miss_set$Q73 > 0] <- 0
miss_set$Q74[miss_set$Q74 > 0] <- 0
miss_set$Q76[miss_set$Q76 > 0] <- 0
miss_set$Q77[miss_set$Q77 > 0] <- 0
miss_set$Q78[miss_set$Q78 > 0] <- 0
miss_set$Q79[miss_set$Q79 > 0] <- 0
miss_set$Q80[miss_set$Q80 > 0] <- 0
miss_set$Q81[miss_set$Q81 > 0] <- 0
miss_set$Q82[miss_set$Q82 > 0] <- 0
miss_set$Q68[is.na(miss_set$Q68)] <- 1
miss_set$Q69[is.na(miss_set$Q69)] <- 1
miss_set$Q70[is.na(miss_set$Q70)] <- 1
miss_set$Q71[is.na(miss_set$Q71)] <- 1
miss_set$Q72[is.na(miss_set$Q72)] <- 1
miss_set$Q73[is.na(miss_set$Q73)] <- 1
miss_set$Q74[is.na(miss_set$Q74)] <- 1
miss_set$Q76[is.na(miss_set$Q76)] <- 1
miss_set$Q77[is.na(miss_set$Q77)] <- 1
```

```r
miss_set$Q78[is.na(miss_set$Q78)] <- 1
miss_set$Q79[is.na(miss_set$Q79)] <- 1
miss_set$Q80[is.na(miss_set$Q80)] <- 1
miss_set$Q81[is.na(miss_set$Q81)] <- 1
miss_set$Q82[is.na(miss_set$Q82)] <- 1

# 3. OBTAIN TOTAL SCORE ON EXTERNAL EVENTS ITEMS
miss_set$total_ext <- rowSums(miss_set[, c(1:52)])  # total score of exposure to abusive ex
miss_set$total_int <- rowSums(miss_set[, c(53:66)])  # a total score of internal states mis

# 4. RUN CORRELATION BETWEEN TOTAL SCORE AND DICHOTOMOUS
# ITEMS
library(psych)
miss_set2 <- miss_set[, c(53:68)]  # delete all but the internal states and total scores
miss_r <- lowerCor(miss_set2)  # the correlation matrix, rounded to 2 decimals
```

```
    Q68   Q69   Q70   Q71   Q72   Q73   Q74   Q76   Q77   Q78   Q79
Q68  1.00
Q69  0.99  1.00
Q70  0.99  0.99  1.00
Q71  1.00  0.99  0.98  1.00
Q72  0.99  0.99  0.99  0.99  1.00
Q73  1.00  0.99  0.98  0.99  0.99  1.00
Q74  0.99  0.98  0.97  0.99  0.98  0.98  1.00
Q76  0.99  0.99  0.98  0.99  0.99  0.99  0.98  1.00
Q77  0.99  0.99  0.98  1.00  0.98  0.99  0.99  0.99  1.00
Q78  0.99  0.99  0.98  0.99  0.98  0.99  0.98  0.98  0.99  1.00
Q79  0.98  0.98  0.97  0.98  0.97  0.98  0.97  0.97  0.98  0.98  1.00
Q80  0.98  0.98  0.98  0.99  0.98  0.98  0.98  0.99  0.98  0.98  0.97
Q81  1.00  0.99  0.99  1.00  0.99  1.00  0.99  0.99  0.99  0.99  0.98
Q82  0.99  0.99  0.98  0.99  0.98  0.98  0.98  0.99  0.99  0.98  0.97
Q84  0.01  0.00  0.00  0.00  0.03  0.03  0.02  0.02    NA  0.01  0.01
Q85  0.01  0.04  0.03  0.01  0.03  0.01 -0.01  0.04    NA  0.06 -0.01
    Q80   Q81   Q82   Q84   Q85
Q80  1.00
Q81  0.98  1.00
Q82  0.98  0.99  1.00
Q84 -0.02  0.01 -0.01  1.00
Q85  0.05  0.01  0.05  0.06  1.00
```

```r
corPlot(miss_r, stars = TRUE, upper = FALSE)  # examine the many options for this function
```

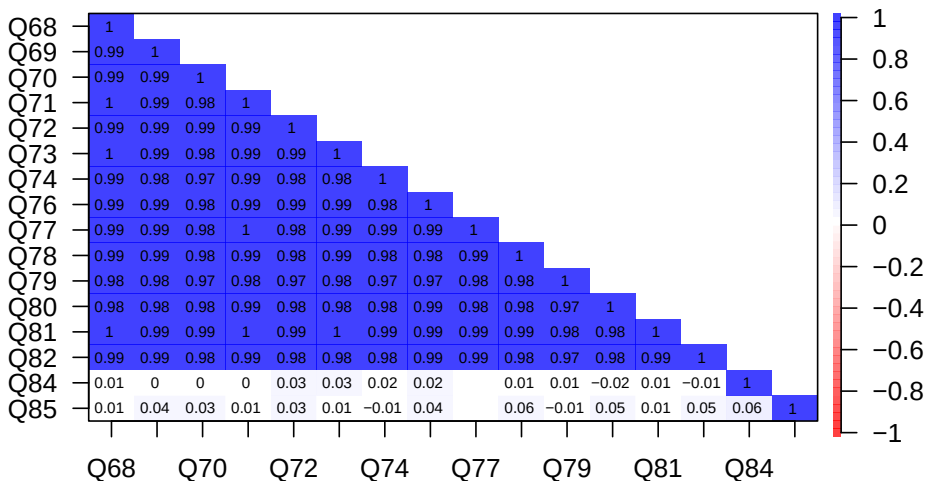


Figure **??** illustrates the correlations among the Internal States items (dichotomized according to their missingness (1=missing, 0=all others)), the total score from the external states (total_ext), and the total score of internal item missingness (total_int). The correlation of total_ext with all the other items is nearly zero, suggesting that more exposure to abusive external states did ***not*** predict nonresponse to the Internal States items. This finding is consistent with the finding from Figure **??** that the missing data appear to missing completely at random.

Missing data methods. There are several different ways to handle missing data. A common method is to *delete all cases missing any data*. This is called **listwise** deletion of data. Its primary advantages are simplicity, expedience, and analysis based on complete information. The disadvantages of listwise are (1) disposal of the information offered by those examinees who responded to some items but not others, and (2) diminishment of the dataset, and with it, a reduction in statistical power. The "all" data file of all Likert items and demographics items derived earlier reports data from 3,222 examinees. The following code chunk creates a subset of complete data. This smaller dataset has 1,841 examinees.

```
# LISTWISE DELETION OF MISSING DATA
all_l <- na.omit(all)  #  Likert items from columns 14-79 plus demographics post listwise
write.csv(all_l, "data/all_l.csv", row.names = FALSE)  # save data

# OTHER POSSIBLE LISTWISE DATASETS

# items <- items[complete.cases(items), ] # create new
# dataset without missing data
items_l <- na.omit(items)  # Likert items from columns 14-79 post listwise
ext_items_l <- na.omit(ext_items)  # Likert items for external events post listwise
int_items_l <- na.omit(int_items)  # Likert items for internal states post listwise
```

```r
write.csv(items_l, "data/items_l.csv", row.names = FALSE)
write.csv(ext_items_l, "data/ext_items_l.csv", row.names = FALSE)
write.csv(int_items_l, "data/int_items_l.csv", row.names = FALSE)
```

Analysis can proceed with these two datasets:

1. the "all" dataset of all Likert and demographic items (N=3,222)
2. the "all_l" dataset of 1,841 examinees who reported complete data

Chapter 6

Item Analysis

In this chapter, we examine the technical quality of the 66 items that Dan piloted for the SHAS. We begin by defining item analysis and discussing its importance. Then document our item analyses from both classical test theory (CTT) and item response theory (IRT) frameworks. We conclude by summarizing what we learn from these analyses to help us refine these items and develop better ones in the future.

6.1 About Item Analysis

```
# LOAD PACKAGES
library(Hmisc)  # for item statistics
library(psych)  # for psychometric analysis
library(psychTools)  # for psychometric analysis
library(CTT)  # for classical test theory stats
library(pastecs)  # for item statistics
library(ltm)  # for point-biserial correlations
library(corrplot)  # for correlation plot
library(knitr)  # how RMarkdown compiles all the code into documents

# LOAD DATA
all <- read.csv("data/all.csv")
items <- read.csv("data/items.csv")
ext_items <- read.csv("data/ext_items.csv")
int_items <- read.csv("data/int_items.csv")
# Datasets with listwise deletion
all_l <- read.csv("data/all_l.csv")
# items_l <- read.csv('data/items_l.csv') ext_items_l <-
# read.csv('data/ext_items_l.csv') int_items_l <-
```

```
# read.csv('data/int_items_l.csv')
```

6.2 Classical Item Analysis

```
# DESCRIBE DATA options(digits=2) describe(items)

# this doesn't work knitr::kable(itm_stats, caption =
# 'Items Statistics', booktabs = TRUE, digits=2)

# A POSSIBLE BAR PLOT OF ITEM STATISTICS
# barplot(item_stats$nbr.val, main='Complete Cases',
# horiz=TRUE, col='red') barplot(item_stats$nbr.val,
# main='Complete Cases', horiz=FALSE, col='red', ylim =
# c(0,8000),names.arg=labs)
```

```
# CLASSICAL ITEM ANALYSIS

# CTT::itemAnalysis(items, itemReport=TRUE, NA.Delete=TRUE,
# rBisML=FALSE, hardFlag, easyFlag, pBisFlag, bisFlag,
# flagStyle = c('X',''))

CTT::itemAnalysis(items, itemReport = TRUE, NA.Delete = TRUE,
    rBisML = FALSE, flagStyle = c("X", ""))
```

```
 Number of Items
 66

 Number of Examinees
 2979

 Coefficient Alpha
 0.981
```

```
# MORE ITEM STATISTICS
library(pastecs)
item_stats <- pastecs::stat.desc(items)  # use the stat.desc() to render data frame of item
item_stats <- t(item_stats)  # transpose the data frame
item_stats <- data.frame(item_stats)
item_stats[order(item_stats$mean), ]  # order by mean
```

```
    nbr.val nbr.null nbr.na min max range   sum median     mean     SE.mean
Q46    3211        0     11   1   5     4  3560      1 1.108689 0.008490661
Q41    3211        0     11   1   5     4  4091      1 1.274058 0.015489036
Q17    3208        0     14   1   5     4  4288      1 1.336658 0.017179617
Q47    3218        0      4   1   5     4  4758      1 1.478558 0.015127455
```

Q60	3214	0	8	1	5	4	4784	1	1.488488	0.017484938
Q55	3205	0	17	1	5	4	5067	1	1.580967	0.016718556
Q72	3070	0	152	1	5	4	5027	1	1.637459	0.017919686
Q81	3073	0	149	1	5	4	5435	1	1.768630	0.018681107
Q65	3220	0	2	1	5	4	5868	1	1.822360	0.020744262
Q10	3220	0	2	1	5	4	5887	1	1.828261	0.020808433
Q26	3212	0	10	1	5	4	6005	1	1.869552	0.022437462
Q24	3221	0	1	1	5	4	6074	1	1.885750	0.020727511
Q23	3219	0	3	1	5	4	6170	1	1.916744	0.019956972
Q27	3217	0	5	1	5	4	6236	1	1.938452	0.023790856
Q31	3217	0	5	1	5	4	6531	2	2.030152	0.020911316
Q30	3219	0	3	1	5	4	6581	2	2.044424	0.022072319
Q43	3217	0	5	1	5	4	6577	2	2.044451	0.020824627
Q68	3073	0	149	1	5	4	6316	2	2.055321	0.022266844
Q52	3217	0	5	1	5	4	6683	2	2.077401	0.022288789
Q70	3069	0	153	1	5	4	6477	2	2.110459	0.021364098
Q49	3213	0	9	1	5	4	6822	2	2.123249	0.021761390
Q16	3217	0	5	1	5	4	6851	2	2.129624	0.022434067
Q57	3219	0	3	1	5	4	6931	2	2.153153	0.023485372
Q20	3217	0	5	1	5	4	6939	2	2.156979	0.024428124
Q62	3220	0	2	1	5	4	6961	2	2.161801	0.021779530
Q59	3217	0	5	1	5	4	6962	2	2.164128	0.021672305
Q21	3218	0	4	1	5	4	7081	1	2.200435	0.025831450
Q50	3219	0	3	1	5	4	7181	1	2.230817	0.025990704
Q82	3071	0	151	1	5	4	6870	2	2.237056	0.021826439
Q64	3220	0	2	1	5	4	7263	2	2.255590	0.022468413
Q28	3216	0	6	1	5	4	7276	2	2.262438	0.022075077
Q40	3220	0	2	1	5	4	7334	2	2.277640	0.024425246
Q63	3217	0	5	1	5	4	7396	2	2.299036	0.021966859
Q19	3219	0	3	1	5	4	7460	2	2.317490	0.023452526
Q51	3216	0	6	1	5	4	7508	2	2.334577	0.024281404
Q53	3218	0	4	1	5	4	7596	2	2.360472	0.023856535
Q79	3068	0	154	1	5	4	7350	2	2.395698	0.022911965
Q34	3219	0	3	1	5	4	7919	2	2.460081	0.023465067
Q29	3213	0	9	1	5	4	7941	2	2.471522	0.023642519
Q73	3072	0	150	1	5	4	7672	2	2.497396	0.023590018
Q11	3216	0	6	1	5	4	8090	2	2.515547	0.023890349
Q58	3220	0	2	1	5	4	8161	2	2.534472	0.023641218
Q7	3215	0	7	1	5	4	8151	2	2.535303	0.025218942
Q42	3215	0	7	1	5	4	8344	3	2.595334	0.022038135
Q44	3217	0	5	1	5	4	8404	3	2.612372	0.023883741
Q56	3217	0	5	1	5	4	8760	3	2.723034	0.024094459
Q18	3220	0	2	1	5	4	8841	3	2.745652	0.024196861
Q45	3219	0	3	1	5	4	8883	3	2.759553	0.025541840
Q78	3071	0	151	1	5	4	8521	3	2.774666	0.023426479
Q36	3216	0	6	1	5	4	8934	3	2.777985	0.023564784

Q61	3219	0	3	1	5	4	8950	3	2.780367	0.025864513
Q32	3219	0	3	1	5	4	8970	3	2.786580	0.023280773
Q74	3071	0	151	1	5	4	8569	3	2.790296	0.023365007
Q76	3072	0	150	1	5	4	8686	3	2.827474	0.026741805
Q54	3217	0	5	1	5	4	9383	3	2.916693	0.024734726
Q37	3221	0	1	1	5	4	9469	3	2.939770	0.024280108
Q38	3217	0	5	1	5	4	9488	3	2.949332	0.024725087
Q71	3074	0	148	1	5	4	9240	3	3.005856	0.024589907
Q69	3071	0	151	1	5	4	9255	3	3.013676	0.021621379
Q39	3219	0	3	1	5	4	9826	3	3.052501	0.026028978
Q80	3070	0	152	1	5	4	9423	3	3.069381	0.022153977
Q77	3075	0	147	1	5	4	9479	3	3.082602	0.023042783
Q35	3217	0	5	1	5	4	10049	3	3.123718	0.024544640
Q25	3217	0	5	1	5	4	10243	3	3.184022	0.023217693
Q66	3216	0	6	1	5	4	10510	3	3.268035	0.024787878
Q22	3218	0	4	1	5	4	10713	4	3.329086	0.026288815

	CI.mean.0.95	var	std.dev	coef.var
Q46	0.01664767	0.2314852	0.4811291	0.4339622
Q41	0.03036940	0.7703517	0.8776968	0.6888987
Q17	0.03368414	0.9468067	0.9730399	0.7279646
Q47	0.02966043	0.7364067	0.8581415	0.5803907
Q60	0.03428276	0.9825939	0.9912587	0.6659502
Q55	0.03278015	0.8958299	0.9464829	0.5986733
Q72	0.03513580	0.9858235	0.9928864	0.6063579
Q81	0.03662873	1.0724271	1.0355805	0.5855270
Q65	0.04067330	1.3856446	1.1771341	0.6459393
Q10	0.04079912	1.3942306	1.1807754	0.6458462
Q26	0.04399320	1.6170483	1.2716321	0.6801803
Q24	0.04064045	1.3838372	1.1763661	0.6238188
Q23	0.03912966	1.2820657	1.1322834	0.5907326
Q27	0.04664678	1.8208375	1.3493841	0.6961143
Q31	0.04100086	1.4067398	1.1860606	0.5842225
Q30	0.04327723	1.5682559	1.2523002	0.6125444
Q43	0.04083089	1.3951006	1.1811438	0.5777314
Q68	0.04365941	1.5236314	1.2343546	0.6005655
Q52	0.04370167	1.5981738	1.2641890	0.6085435
Q70	0.04188939	1.4007674	1.1835402	0.5607974
Q49	0.04266762	1.5215421	1.2335081	0.5809530
Q16	0.04398652	1.6190755	1.2724290	0.5974900
Q57	0.04604780	1.7754803	1.3324715	0.6188466
Q20	0.04789627	1.9196909	1.3855291	0.6423472
Q62	0.04270315	1.5274003	1.2358804	0.5716901
Q59	0.04249293	1.5109889	1.2292229	0.5679991
Q21	0.05064777	2.1472550	1.4653515	0.6659372
Q50	0.05096001	2.1744882	1.4746146	0.6610200
Q82	0.04279591	1.4630042	1.2095471	0.5406869

```
Q64   0.04405384 1.6255512 1.2749711 0.5652495
Q28   0.04328265 1.5671858 1.2518729 0.5533292
Q40   0.04789061 1.9210283 1.3860116 0.6085298
Q63   0.04307046 1.5523405 1.2459296 0.5419356
Q19   0.04598340 1.7705175 1.3306079 0.5741591
Q51   0.04760860 1.8961104 1.3769933 0.5898256
Q53   0.04677555 1.8314741 1.3533196 0.5733258
Q79   0.04492436 1.6105716 1.2690830 0.5297342
Q34   0.04600799 1.7724116 1.3313195 0.5411690
Q29   0.04635595 1.7959665 1.3401367 0.5422314
Q73   0.04625382 1.7095341 1.3074915 0.5235420
Q11   0.04684186 1.8355280 1.3548166 0.5385773
Q58   0.04635336 1.7996811 1.3415220 0.5293102
Q7    0.04944684 2.0447240 1.4299385 0.5640108
Q42   0.04321022 1.5614592 1.2495836 0.4814731
Q44   0.04682890 1.8350832 1.3546524 0.5185527
Q56   0.04724205 1.8676067 1.3666041 0.5018682
Q18   0.04744281 1.8852716 1.3730519 0.5000822
Q45   0.05007992 2.1000291 1.4491477 0.5251386
Q78   0.04593316 1.6853645 1.2982159 0.4678818
Q36   0.04620352 1.7858418 1.3363539 0.4810515
Q61   0.05071259 2.1534242 1.4674550 0.5277919
Q32   0.04564665 1.7446800 1.3208634 0.4740088
Q74   0.04581263 1.6765312 1.2948093 0.4640401
Q76   0.05243364 2.1968613 1.4821813 0.5242069
Q54   0.04849742 1.9681821 1.4029191 0.4809966
Q37   0.04760603 1.8988557 1.3779897 0.4687406
Q38   0.04847852 1.9666483 1.4023724 0.4754882
Q71   0.04821432 1.8587356 1.3633546 0.4535662
Q69   0.04239384 1.4356435 1.1981834 0.3975820
Q39   0.05103505 2.1808973 1.4767861 0.4837955
Q80   0.04343813 1.5067520 1.2274983 0.3999172
Q77   0.04518081 1.6327323 1.2777841 0.4145148
Q35   0.04812472 1.9380474 1.3921377 0.4456669
Q25   0.04552297 1.7341601 1.3168751 0.4135885
Q66   0.04860164 1.9760355 1.4057153 0.4301409
Q22   0.05154452 2.2239657 1.4912966 0.4479597
```

```r
item_stats[order(item_stats$nbr.val), ]  # order by N cases
```

```
    nbr.val nbr.null nbr.na min max range   sum median     mean     SE.mean
Q79    3068        0    154   1   5     4  7350      2 2.395698 0.022911965
Q70    3069        0    153   1   5     4  6477      2 2.110459 0.021364098
Q72    3070        0    152   1   5     4  5027      1 1.637459 0.017919686
Q80    3070        0    152   1   5     4  9423      3 3.069381 0.022153977
Q69    3071        0    151   1   5     4  9255      3 3.013676 0.021621379
```

Q74	3071	0	151	1	5	4	8569	3	2.790296	0.023365007
Q78	3071	0	151	1	5	4	8521	3	2.774666	0.023426479
Q82	3071	0	151	1	5	4	6870	2	2.237056	0.021826439
Q73	3072	0	150	1	5	4	7672	2	2.497396	0.023590018
Q76	3072	0	150	1	5	4	8686	3	2.827474	0.026741805
Q68	3073	0	149	1	5	4	6316	2	2.055321	0.022266844
Q81	3073	0	149	1	5	4	5435	1	1.768630	0.018681107
Q71	3074	0	148	1	5	4	9240	3	3.005856	0.024589907
Q77	3075	0	147	1	5	4	9479	3	3.082602	0.023042783
Q55	3205	0	17	1	5	4	5067	1	1.580967	0.016718556
Q17	3208	0	14	1	5	4	4288	1	1.336658	0.017179617
Q41	3211	0	11	1	5	4	4091	1	1.274058	0.015489036
Q46	3211	0	11	1	5	4	3560	1	1.108689	0.008490661
Q26	3212	0	10	1	5	4	6005	1	1.869552	0.022437462
Q29	3213	0	9	1	5	4	7941	2	2.471522	0.023642519
Q49	3213	0	9	1	5	4	6822	2	2.123249	0.021761390
Q60	3214	0	8	1	5	4	4784	1	1.488488	0.017484938
Q7	3215	0	7	1	5	4	8151	2	2.535303	0.025218942
Q42	3215	0	7	1	5	4	8344	3	2.595334	0.022038135
Q11	3216	0	6	1	5	4	8090	2	2.515547	0.023890349
Q28	3216	0	6	1	5	4	7276	2	2.262438	0.022075077
Q36	3216	0	6	1	5	4	8934	3	2.777985	0.023564784
Q51	3216	0	6	1	5	4	7508	2	2.334577	0.024281404
Q66	3216	0	6	1	5	4	10510	3	3.268035	0.024787878
Q16	3217	0	5	1	5	4	6851	2	2.129624	0.022434067
Q20	3217	0	5	1	5	4	6939	2	2.156979	0.024428124
Q25	3217	0	5	1	5	4	10243	3	3.184022	0.023217693
Q27	3217	0	5	1	5	4	6236	1	1.938452	0.023790856
Q31	3217	0	5	1	5	4	6531	2	2.030152	0.020911316
Q35	3217	0	5	1	5	4	10049	3	3.123718	0.024544640
Q38	3217	0	5	1	5	4	9488	3	2.949332	0.024725087
Q43	3217	0	5	1	5	4	6577	2	2.044451	0.020824627
Q44	3217	0	5	1	5	4	8404	3	2.612372	0.023883741
Q52	3217	0	5	1	5	4	6683	2	2.077401	0.022288789
Q54	3217	0	5	1	5	4	9383	3	2.916693	0.024734726
Q56	3217	0	5	1	5	4	8760	3	2.723034	0.024094459
Q59	3217	0	5	1	5	4	6962	2	2.164128	0.021672305
Q63	3217	0	5	1	5	4	7396	2	2.299036	0.021966859
Q21	3218	0	4	1	5	4	7081	1	2.200435	0.025831450
Q22	3218	0	4	1	5	4	10713	4	3.329086	0.026288815
Q47	3218	0	4	1	5	4	4758	1	1.478558	0.015127455
Q53	3218	0	4	1	5	4	7596	2	2.360472	0.023856535
Q19	3219	0	3	1	5	4	7460	2	2.317490	0.023452526
Q23	3219	0	3	1	5	4	6170	1	1.916744	0.019956972
Q30	3219	0	3	1	5	4	6581	2	2.044424	0.022072319
Q32	3219	0	3	1	5	4	8970	3	2.786580	0.023280773

Q34	3219	0	3	1	5	4	7919	2	2.460081	0.023465067
Q39	3219	0	3	1	5	4	9826	3	3.052501	0.026028978
Q45	3219	0	3	1	5	4	8883	3	2.759553	0.025541840
Q50	3219	0	3	1	5	4	7181	1	2.230817	0.025990704
Q57	3219	0	3	1	5	4	6931	2	2.153153	0.023485372
Q61	3219	0	3	1	5	4	8950	3	2.780367	0.025864513
Q10	3220	0	2	1	5	4	5887	1	1.828261	0.020808433
Q18	3220	0	2	1	5	4	8841	3	2.745652	0.024196861
Q40	3220	0	2	1	5	4	7334	2	2.277640	0.024425246
Q58	3220	0	2	1	5	4	8161	2	2.534472	0.023641218
Q62	3220	0	2	1	5	4	6961	2	2.161801	0.021779530
Q64	3220	0	2	1	5	4	7263	2	2.255590	0.022468413
Q65	3220	0	2	1	5	4	5868	1	1.822360	0.020744262
Q24	3221	0	1	1	5	4	6074	1	1.885750	0.020727511
Q37	3221	0	1	1	5	4	9469	3	2.939770	0.024280108

	CI.mean.0.95	var	std.dev	coef.var
Q79	0.04492436	1.6105716	1.2690830	0.5297342
Q70	0.04188939	1.4007674	1.1835402	0.5607974
Q72	0.03513580	0.9858235	0.9928864	0.6063579
Q80	0.04343813	1.5067520	1.2274983	0.3999172
Q69	0.04239384	1.4356435	1.1981834	0.3975820
Q74	0.04581263	1.6765312	1.2948093	0.4640401
Q78	0.04593316	1.6853645	1.2982159	0.4678818
Q82	0.04279591	1.4630042	1.2095471	0.5406869
Q73	0.04625382	1.7095341	1.3074915	0.5235420
Q76	0.05243364	2.1968613	1.4821813	0.5242069
Q68	0.04365941	1.5236314	1.2343546	0.6005655
Q81	0.03662873	1.0724271	1.0355805	0.5855270
Q71	0.04821432	1.8587356	1.3633546	0.4535662
Q77	0.04518081	1.6327323	1.2777841	0.4145148
Q55	0.03278015	0.8958299	0.9464829	0.5986733
Q17	0.03368414	0.9468067	0.9730399	0.7279646
Q41	0.03036940	0.7703517	0.8776968	0.6888987
Q46	0.01664767	0.2314852	0.4811291	0.4339622
Q26	0.04399320	1.6170483	1.2716321	0.6801803
Q29	0.04635595	1.7959665	1.3401367	0.5422314
Q49	0.04266762	1.5215421	1.2335081	0.5809530
Q60	0.03428276	0.9825939	0.9912587	0.6659502
Q7	0.04944684	2.0447240	1.4299385	0.5640108
Q42	0.04321022	1.5614592	1.2495836	0.4814731
Q11	0.04684186	1.8355280	1.3548166	0.5385773
Q28	0.04328265	1.5671858	1.2518729	0.5533292
Q36	0.04620352	1.7858418	1.3363539	0.4810515
Q51	0.04760860	1.8961104	1.3769933	0.5898256
Q66	0.04860164	1.9760355	1.4057153	0.4301409
Q16	0.04398652	1.6190755	1.2724290	0.5974900

```
Q20   0.04789627 1.9196909 1.3855291 0.6423472
Q25   0.04552297 1.7341601 1.3168751 0.4135885
Q27   0.04664678 1.8208375 1.3493841 0.6961143
Q31   0.04100086 1.4067398 1.1860606 0.5842225
Q35   0.04812472 1.9380474 1.3921377 0.4456669
Q38   0.04847852 1.9666483 1.4023724 0.4754882
Q43   0.04083089 1.3951006 1.1811438 0.5777314
Q44   0.04682890 1.8350832 1.3546524 0.5185527
Q52   0.04370167 1.5981738 1.2641890 0.6085435
Q54   0.04849742 1.9681821 1.4029191 0.4809966
Q56   0.04724205 1.8676067 1.3666041 0.5018682
Q59   0.04249293 1.5109889 1.2292229 0.5679991
Q63   0.04307046 1.5523405 1.2459296 0.5419356
Q21   0.05064777 2.1472550 1.4653515 0.6659372
Q22   0.05154452 2.2239657 1.4912966 0.4479597
Q47   0.02966043 0.7364067 0.8581415 0.5803907
Q53   0.04677555 1.8314741 1.3533196 0.5733258
Q19   0.04598340 1.7705175 1.3306079 0.5741591
Q23   0.03912966 1.2820657 1.1322834 0.5907326
Q30   0.04327723 1.5682559 1.2523002 0.6125444
Q32   0.04564665 1.7446800 1.3208634 0.4740088
Q34   0.04600799 1.7724116 1.3313195 0.5411690
Q39   0.05103505 2.1808973 1.4767861 0.4837955
Q45   0.05007992 2.1000291 1.4491477 0.5251386
Q50   0.05096001 2.1744882 1.4746146 0.6610200
Q57   0.04604780 1.7754803 1.3324715 0.6188466
Q61   0.05071259 2.1534242 1.4674550 0.5277919
Q10   0.04079912 1.3942306 1.1807754 0.6458462
Q18   0.04744281 1.8852716 1.3730519 0.5000822
Q40   0.04789061 1.9210283 1.3860116 0.6085298
Q58   0.04635336 1.7996811 1.3415220 0.5293102
Q62   0.04270315 1.5274003 1.2358804 0.5716901
Q64   0.04405384 1.6255512 1.2749711 0.5652495
Q65   0.04067330 1.3856446 1.1771341 0.6459393
Q24   0.04064045 1.3838372 1.1763661 0.6238188
Q37   0.04760603 1.8988557 1.3779897 0.4687406
```

Item-Total Correlations.

```
# POINT BISERIAL CORRELATIONS ltm::biserial.cor(x, y, use =
# c('all.obs', 'complete.obs'), level = 1)
```

Distractor Analysis.

```
# DISTRACTOR ANALYSIS CTT::distractorAnalysis
```

Outlier Analysis.

```r
# OUTLIERS psych::outlier(items)
```

Skew and Kurtosis.

6.3 Item analyses in item response theory (IRT) framework

One outcome of this analysis is insight into whether items have too many response categories.

This ltm package "provides a flexible framework for Item Response Theory analyses for dichotomous and polytomous data under a Marginal Maximum Likelihood approach. The fitting algorithms provide valid inferences under Missing At Random missing data mechanisms" [**?**].

```r
# DESCRIPTIVES
ltm::descript(items)
```

```
Descriptive statistics for the 'items' data-set

Sample:
 66 items and 3222 sample units; 2373 missing values

Proportions for each level of response:
         1      2      3      4      5
Q7  0.3521 0.1760 0.1863 0.1555 0.1300
Q10 0.5894 0.1568 0.1339 0.0758 0.0441
Q11 0.3234 0.2086 0.1953 0.1744 0.0983
Q16 0.4492 0.2111 0.1654 0.1097 0.0647
Q17 0.8756 0.0218 0.0324 0.0305 0.0396
Q18 0.2693 0.1655 0.2441 0.1925 0.1286
Q19 0.3995 0.1839 0.1929 0.1469 0.0767
Q20 0.4974 0.1498 0.1458 0.1125 0.0945
Q21 0.5103 0.1346 0.1271 0.1007 0.1274
Q22 0.1961 0.1140 0.1535 0.2374 0.2989
Q23 0.5107 0.2057 0.1712 0.0811 0.0314
Q24 0.5567 0.1608 0.1614 0.0823 0.0388
Q25 0.1532 0.1470 0.2471 0.2676 0.1850
Q26 0.6043 0.1370 0.1062 0.0900 0.0626
Q27 0.6049 0.1066 0.1166 0.0889 0.0830
Q28 0.3741 0.2422 0.1906 0.1334 0.0597
Q29 0.3402 0.1911 0.2157 0.1631 0.0899
Q30 0.4995 0.1687 0.1705 0.1103 0.0509
Q31 0.4629 0.2247 0.1769 0.0905 0.0451
Q32 0.2423 0.1637 0.2700 0.2131 0.1109
Q34 0.3395 0.1948 0.2199 0.1575 0.0882
```

Q35	0.1878	0.1442	0.2316	0.2294	0.2070
Q36	0.2385	0.1931	0.2400	0.2086	0.1197
Q37	0.2155	0.1704	0.2372	0.2127	0.1642
Q38	0.2238	0.1607	0.2353	0.2027	0.1775
Q39	0.2377	0.1280	0.2029	0.2072	0.2243
Q40	0.4382	0.1661	0.1770	0.1171	0.1016
Q41	0.8938	0.0240	0.0287	0.0215	0.0321
Q42	0.2470	0.2439	0.2560	0.1733	0.0799
Q43	0.4563	0.2216	0.1843	0.0967	0.0410
Q44	0.3028	0.1769	0.2266	0.1927	0.1010
Q45	0.2870	0.1730	0.1979	0.1774	0.1646
Q46	0.9396	0.0271	0.0224	0.0069	0.0040
Q47	0.7026	0.1659	0.0942	0.0249	0.0124
Q49	0.4367	0.2210	0.1771	0.1130	0.0523
Q50	0.5228	0.0845	0.1460	0.1323	0.1143
Q51	0.4070	0.1853	0.1692	0.1430	0.0955
Q52	0.4815	0.1871	0.1579	0.1194	0.0541
Q53	0.3875	0.1858	0.1967	0.1386	0.0914
Q54	0.2331	0.1595	0.2350	0.2024	0.1700
Q55	0.6559	0.1807	0.1061	0.0415	0.0159
Q56	0.2729	0.1747	0.2257	0.2098	0.1169
Q57	0.4772	0.1631	0.1640	0.1208	0.0749
Q58	0.3193	0.1845	0.2373	0.1606	0.0984
Q59	0.4200	0.2148	0.1943	0.1231	0.0479
Q60	0.7570	0.0949	0.0741	0.0507	0.0233
Q61	0.2961	0.1528	0.1951	0.1867	0.1693
Q62	0.4317	0.1891	0.2146	0.1149	0.0497
Q63	0.3593	0.2338	0.2157	0.1309	0.0603
Q64	0.3879	0.2286	0.1932	0.1208	0.0696
Q65	0.5891	0.1640	0.1258	0.0776	0.0435
Q66	0.1760	0.1191	0.2083	0.2540	0.2425
Q68	0.4878	0.1705	0.1865	0.1090	0.0462
Q69	0.1309	0.1960	0.3237	0.2273	0.1221
Q70	0.4190	0.2369	0.2053	0.0922	0.0466
Q71	0.1932	0.1724	0.2437	0.2167	0.1740
Q72	0.6375	0.1700	0.1300	0.0427	0.0199
Q73	0.3118	0.2083	0.2386	0.1530	0.0882
Q74	0.2234	0.1869	0.2735	0.2084	0.1078
Q76	0.2839	0.1589	0.1921	0.1764	0.1888
Q77	0.1502	0.1782	0.2595	0.2628	0.1493
Q78	0.2263	0.1941	0.2657	0.2064	0.1075
Q79	0.3286	0.2262	0.2441	0.1232	0.0779
Q80	0.1391	0.1762	0.2883	0.2691	0.1274
Q81	0.5600	0.2089	0.1542	0.0560	0.0208
Q82	0.3751	0.2250	0.2410	0.1055	0.0534

Missing responses:

	Q7	Q10	Q11	Q16	Q17	Q18	Q19	Q20	Q21	Q22
Freq	7.0000	2.0000	6.0000	5.0000	14.0000	2.0000	3.0000	5.0000	4.0000	4.0000
(%)	0.2173	0.0621	0.1862	0.1552	0.4345	0.0621	0.0931	0.1552	0.1241	0.1241

	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32
Freq	3.0000	1.000	5.0000	10.0000	5.0000	6.0000	9.0000	3.0000	5.0000	3.0000
(%)	0.0931	0.031	0.1552	0.3104	0.1552	0.1862	0.2793	0.0931	0.1552	0.0931

	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43
Freq	3.0000	5.0000	6.0000	1.000	5.0000	3.0000	2.0000	11.0000	7.0000	5.0000
(%)	0.0931	0.1552	0.1862	0.031	0.1552	0.0931	0.0621	0.3414	0.2173	0.1552

	Q44	Q45	Q46	Q47	Q49	Q50	Q51	Q52	Q53	Q54
Freq	5.0000	3.0000	11.0000	4.0000	9.0000	3.0000	6.0000	5.0000	4.0000	5.0000
(%)	0.1552	0.0931	0.3414	0.1241	0.2793	0.0931	0.1862	0.1552	0.1241	0.1552

	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64
Freq	17.0000	5.0000	3.0000	2.0000	5.0000	8.0000	3.0000	2.0000	5.0000	2.0000
(%)	0.5276	0.1552	0.0931	0.0621	0.1552	0.2483	0.0931	0.0621	0.1552	0.0621

	Q65	Q66	Q68	Q69	Q70	Q71	Q72	Q73
Freq	2.0000	6.0000	149.0000	151.0000	153.0000	148.0000	152.0000	150.0000
(%)	0.0621	0.1862	4.6245	4.6865	4.7486	4.5934	4.7176	4.6555

	Q74	Q76	Q77	Q78	Q79	Q80	Q81	Q82
Freq	151.0000	150.0000	147.0000	151.0000	154.0000	152.0000	149.0000	151.0000
(%)	4.6865	4.6555	4.5624	4.6865	4.7796	4.7176	4.6245	4.6865

Frequencies of total scores:

	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Freq	30	12	14	21	17	25	16	17	13	18	22	14	15	20	20	15	30	16	12	16	20	18	24	24	17
	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111				
Freq	22	34	17	25	20	24	13	25	16	16	18	21	15	17	23	23	23	21	17	18	17				
	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129							
Freq	20	16	15	21	21	16	17	21	18	22	20	19	24	13	14	19	17	14							
	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147							
Freq	16	19	17	12	16	18	19	23	13	18	15	18	19	16	21	14	26	17							
	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165							
Freq	19	17	16	16	14	27	18	15	27	19	23	28	9	11	30	27	16	14							
	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183							
Freq	26	15	23	10	17	16	11	16	14	18	13	13	18	14	14	18	21	16							
	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201							
Freq	17	17	10	13	16	17	17	13	13	19	22	17	18	12	13	9	15	25							
	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219							
Freq	15	16	14	14	14	12	22	10	13	20	19	18	17	7	12	10	17	8							
	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237							
Freq	12	7	18	14	15	13	7	11	14	11	8	11	9	14	13	8	11	7							
	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255							
Freq	8	8	6	9	6	8	10	7	8	7	3	9	6	7	5	8	7	7							
	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273							

```
Freq   4   3   5   6   7   5   8   2   3  11   4   3   4   7   1   3   5   4
     274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291
Freq   1   2   3   3   3   3   3   2   2   2   1   0   2   2   3   2   3   1
     292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309
Freq   0   2   1   1   1   2   2   1   0   1   2   0   0   0   0   0   0   0
     310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327
Freq   2   0   0   2   0   0   0   0   0   0   1   0   0   0   0   0   0   0
     328 329 330
Freq   0   0   0


Cronbach's alpha:
               value
All Items     0.9807
Excluding Q7  0.9804
Excluding Q10 0.9805
Excluding Q11 0.9802
Excluding Q16 0.9804
Excluding Q17 0.9808
Excluding Q18 0.9802
Excluding Q19 0.9803
Excluding Q20 0.9804
Excluding Q21 0.9804
Excluding Q22 0.9804
Excluding Q23 0.9805
Excluding Q24 0.9804
Excluding Q25 0.9803
Excluding Q26 0.9805
Excluding Q27 0.9807
Excluding Q28 0.9803
Excluding Q29 0.9804
Excluding Q30 0.9804
Excluding Q31 0.9804
Excluding Q32 0.9803
Excluding Q34 0.9805
Excluding Q35 0.9803
Excluding Q36 0.9803
Excluding Q37 0.9803
Excluding Q38 0.9802
Excluding Q39 0.9803
Excluding Q40 0.9803
Excluding Q41 0.9809
Excluding Q42 0.9803
Excluding Q43 0.9803
Excluding Q44 0.9802
Excluding Q45 0.9803
```

```
Excluding Q46 0.9810
Excluding Q47 0.9806
Excluding Q49 0.9803
Excluding Q50 0.9807
Excluding Q51 0.9805
Excluding Q52 0.9803
Excluding Q53 0.9803
Excluding Q54 0.9802
Excluding Q55 0.9809
Excluding Q56 0.9802
Excluding Q57 0.9803
Excluding Q58 0.9803
Excluding Q59 0.9803
Excluding Q60 0.9807
Excluding Q61 0.9803
Excluding Q62 0.9804
Excluding Q63 0.9803
Excluding Q64 0.9804
Excluding Q65 0.9805
Excluding Q66 0.9804
Excluding Q68 0.9804
Excluding Q69 0.9806
Excluding Q70 0.9806
Excluding Q71 0.9803
Excluding Q72 0.9806
Excluding Q73 0.9805
Excluding Q74 0.9805
Excluding Q76 0.9803
Excluding Q77 0.9802
Excluding Q78 0.9804
Excluding Q79 0.9805
Excluding Q80 0.9804
Excluding Q81 0.9805
Excluding Q82 0.9805


Pairwise Associations:
   Item i Item j p.value
1      33     62   0.125
2      33     54   0.033
3      33     63   0.027
4      33     60   0.020
5      22     33   0.019
6      33     48   0.019
7       6     33   0.016
8      33     34   0.015
```

```
9       33      53   0.015
10      33      58   0.013
```

6.3.1 Item thresholds

6.3.2 Item characteristic curves

6.3.3 Differential item functioning (DIF)

6.4 Summary

Chapter 7

Reliability

In this chapter, we addresses the reliability of the SHAS based on the 66 piloted items. We begin by defining reliability and discussing its importance. Then we report the results of our analyses of the reliability of the piloted SHAS items from both classical test theory (CTT) and item response theory (IRT) frameworks. Since these are cross-sectional data taken at one point in time, our work is limited to *internal consistency* reliability which focuses on the relationships between the items at one point in time rather than relationships between items or scores between two or more time points. We conclude by considering what this information contributes to improved SHAS instrumentation.

7.1 About reliability

Reliability is the extent to which items and their scores are consistently measuring the trait of interest. It is evidence that two examinees who provide the same responses to the same items, and by extension receive the same total scores, truly have the same level or amount of, or exposure to, the trait.

7.2 Reliability in Classical Test Theory (CTT)

One way to think about reliability in classical test theory is the ratio of signal to noise in the observed data. Do the various item responses and scores provide a true sort of examinees on the trait, or is all this variation merely noise?

The foundation of internal consistency reliability in CTT is relationships between item responses: correlations. Strong correlations among the items are evidence that the items are measuring the same construct. Thus we begin by examining correlation matrices.

```r
# HEATMAP OF CORRELATIONS
psych::corPlot(ext_items_l, upper = FALSE)
```

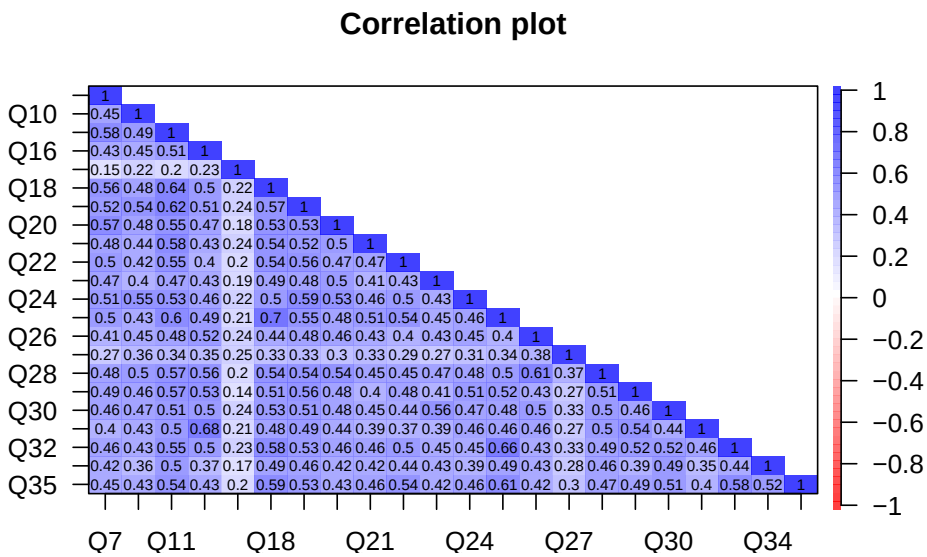


Figure 7.1: Correlation Matrix of External Events Items

Figure 7.1 is a correlation matrix of the External Events items for the *3,174* examinees who responded to all of these items. Each cell reports the correlation coefficient between scores of the two items. All of the cells are color coded to the scale of correlations which range from 0 (indicating no relationship) to 1 (indicating a perfect linear relationship). The cells along the diagonal are all 1.0 because they represent the correlation of the item with itself. Only one half of the matrix is necessary to report all the correlations among all the items.

The plot offers several interesting pieces of information. The first is that Q17–"Being treated as 'less than' because of my sexual orientation–does not quite fit. Responses to this item correlate weakly with responses to all the other items. It is possible that so few examinees experience this event that the responses vary very little, which would attenuate any correlation. Or it may be measuring something different than what the other items in this set. Similar could be said of Q27–"Being denied opportunities to serve/volunteer because of my gender."

```r
# HEATMAP OF CORRELATIONS
psych::corPlot(int_items_l, upper = FALSE)
```

Figure 7.2 is a correlation matrix of the Internal States items for the *3,051* examinees who responded to all of these items. This plot is interesting in that most of the correlations are in the moderate (0.50) range. They are sorting examinees in a similar way, but together they do not measure exactly the same

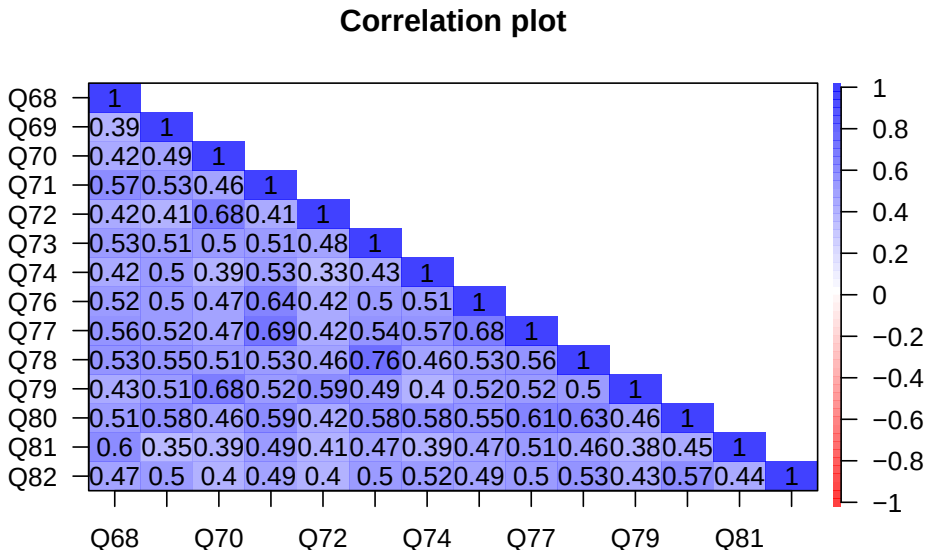


Figure 7.2: Correlation Matrix of Internal States Items

thing.

```
# HEATMAP OF CORRELATIONS
psych::corPlot(items_l, upper = FALSE)
```

This final matrix, Figure 7.3, reports the correlations among all of the Likert items among the *2,979* examinees who responded to all 66 items. Note that missing data are a challenge here, as these plots require complete data.

This plot offers several interesting insights. First, the preponderance of blue indicates that everything is positively correlated. This makes some sense because the items were worded and all the response categories were coded in the same direction.

Second, in this plot there are too many items and correlation coefficients in one place to be distinguishable, but with the colors it is possible to see an overall pattern. This plot orders the items by their order on the questionnaire so we can see how the External Events items correlate with the Internal Events items: moderately. It is possible that these are not two different scales.

Finally, we see at a glance a handful of items that correlate very weakly with all the other items. These items should be scrutinized for refinement and possible omission from the analysis.

Cronbach's coefficient alpha. Arguably the most well-known statistic for estimating internal consistency reliability is Cronbach's coefficient alpha (citation). This is a measure of the average inter-item correlation (?). It is also a measure of the ratio of signal to noise (X = T + e). A ballpark cutoff value is

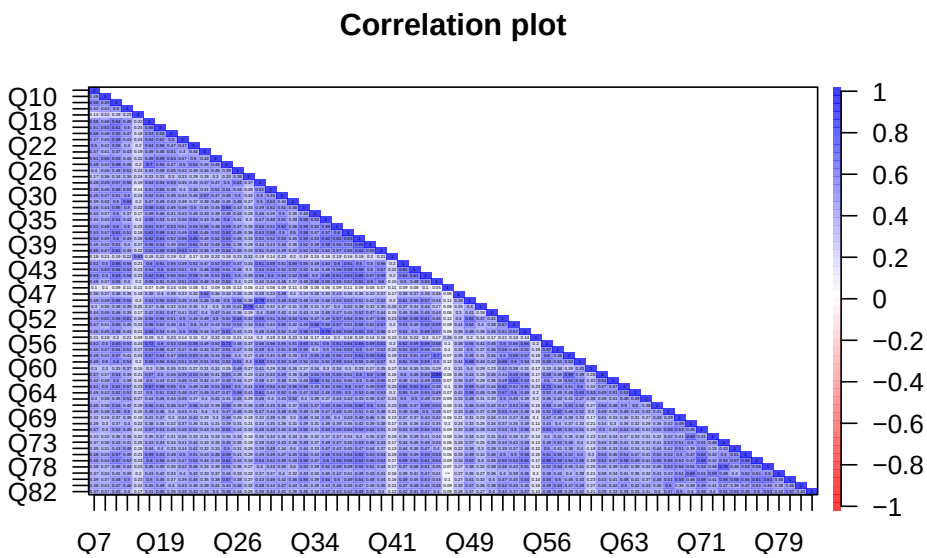


Figure 7.3: Correlation Matrix of All 66 Likert SHAS Items

0.70 [Nunnally and Bernstein, 1994].

```
# CRONBACH'S ALPHA
reliability(items)  # , itemal = TRUE, NA.Delete = TRUE, ml = TRUE)
```

```
Number of Items
66

Number of Examinees
2979

Coefficient Alpha
0.981
```

This output is not pretty, but it does report a Cronbach's alpha of 0.98. This is a high level of internal consistency reliability.

7.3 Reliability in Item Response Theory (IRT)

Wright and Masters (1982) developed reliability measures based on the Rasch measurement model, which have a different concept from classical test theory (e.g., Cronbach's alpha). Reliability is estimated both for persons and items, by means of Person Separation Reliability (PSR). This is an estimate of how well this instrument can distinguish respondents on the measured variable. In parallel, Item Separation Reliability (ISR) is an indication of how well items are separated by the persons taking the test (Wright & Stone, 1999). The cutoff of

separation reliabilities is > 0.80 (Bond & Fox, 2015).

7.4 Summary

Chapter 8

Dimensionality

This chapter deals with the topic of dimensionality. It begins with a definition of dimensionality and explains how it is important. It outlines how to investigate it and suggests what counts as good evidence. Then it reports our study of the dimensionality of the SHAS.

8.1 About Dimensionality

Dimensionality is the study of the dimensions (also known as "factors", "components", and "latent traits") that the items are designed and assembled to collectively measure. This study tends to take two forms. One is exploration of what dimensions a set of items apparently measure, regardless of what they intended to measure. The other is what evidence exist to confirm that a set of items actually measure the dimension they are intended to measure.

Dimensionality is important for the reporting of scores. Scores, as summary measures of a set of items, are only meaningful to the extent that they enjoy evidence that they measure the dimension(s) they are designed to measure. Scores, in turn, are important to the extent that they support decisions that have consequences in the real world. Scores are needed to decide who gets to practice medicine, or fly an airplane.

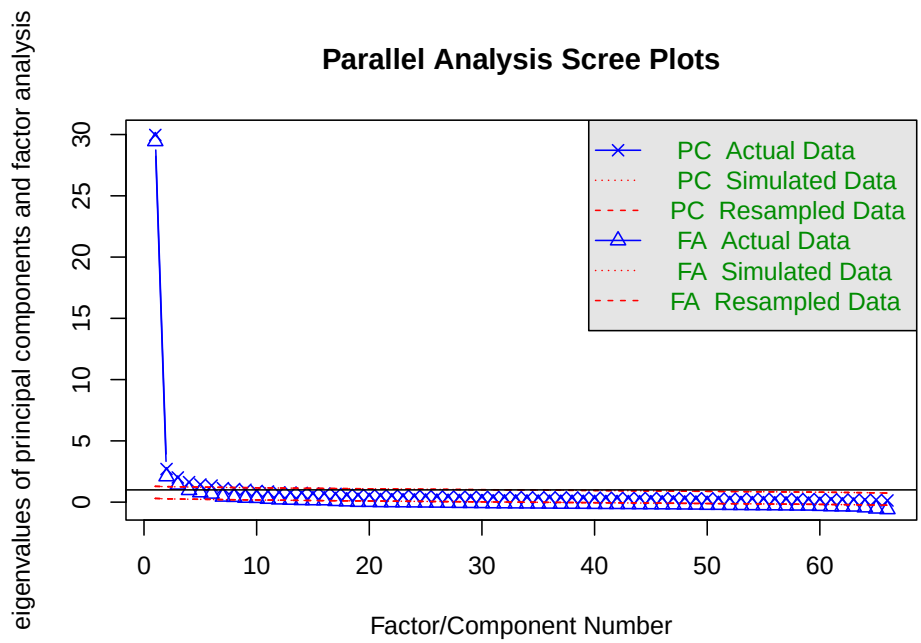
Dimensionality is an important part of the validation process for psychological measures [Embretson and Reise, 2000].

The methodology commonly used to investigate dimensionality is factor analysis. To assess the dimensionality of these 66 piloted SHAS items we use exploratory factor analysis and confirmatory factor analysis (CFA) and PCAR.

8.2 Exploratory Factor Analysis (EFA)

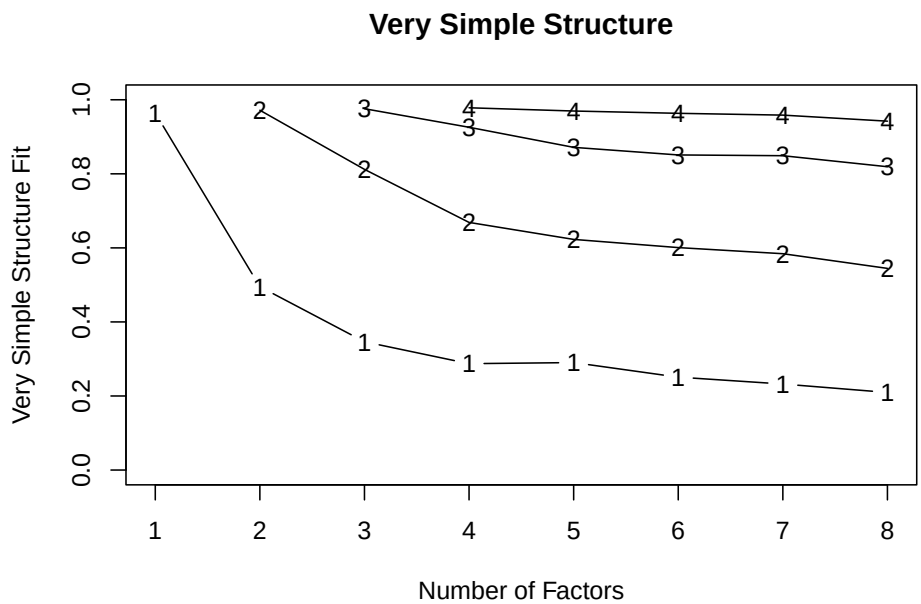
```
# LOAD PACKAGES
library(lavaan)
library(psych)
library(psychTools)

# LOAD DATA all <- read.csv('data/all.csv')
items <- read.csv("data/items.csv")
ext_items <- read.csv("data/ext_items.csv")
int_items <- read.csv("data/int_items.csv")
```

```
# TEST FOR NUMBER OF FACTORS IN DATA USING PARALLEL
# ANALYSIS AND VERY SIMPLE STRUCTURE (VSS)
fa.parallel(items)
```




```
Parallel analysis suggests that the number of factors =  16  and the number of components =
```

```
vss(items)
```




```
Very Simple Structure
Call: vss(x = items)
VSS complexity 1 achieves a maximimum of 0.96  with  1  factors
VSS complexity 2 achieves a maximimum of 0.97  with  2  factors

The Velicer MAP achieves a minimum of 0.01  with  6  factors
BIC achieves a minimum of  -788.85  with  8  factors
Sample Size adjusted BIC achieves a minimum of  4438.02  with  8  factors

Statistics by number of factors
  vss1 vss2    map  dof chisq prob sqresid  fit RMSEA   BIC SABIC complex
1 0.96 0.00 0.0091 2079 37231    0      33 0.96 0.072 20438 27044     1.0
2 0.50 0.97 0.0068 2014 29480    0      26 0.97 0.065 13211 19611     1.6
3 0.35 0.81 0.0058 1950 24035    0      23 0.98 0.059  8283 14479     2.1
4 0.29 0.67 0.0056 1887 20947    0      20 0.98 0.056  5704 11700     2.5
5 0.29 0.62 0.0055 1825 17804    0      18 0.98 0.052  3062  8861     2.7
6 0.25 0.60 0.0052 1764 15824    0      16 0.98 0.050  1575  7180     2.8
7 0.23 0.58 0.0056 1704 13729    0      15 0.98 0.047   -35  5379     2.9
8 0.21 0.54 0.0055 1645 12499    0      14 0.98 0.045  -789  4438     3.0
  eChisq  SRMR eCRMS  eBIC
1  37967 0.052 0.053 21173
2  23187 0.041 0.042  6918
3  16097 0.034 0.036   345
4  12506 0.030 0.032 -2737
5   9591 0.026 0.029 -5151
6   7329 0.023 0.025 -6921
```

```
7   5949 0.021 0.023 -7815
8   5053 0.019 0.022 -8235
```

```r
# PRINCIPAL COMPONENTS ANALYSIS
principal(items)
```

```
Principal Components Analysis
Call: principal(r = items)
Standardized loadings (pattern matrix) based upon correlation matrix
     PC1    h2   u2 com
Q7  0.67 0.451 0.55   1
Q10 0.64 0.404 0.60   1
Q11 0.77 0.598 0.40   1
Q16 0.69 0.481 0.52   1
Q17 0.33 0.112 0.89   1
Q18 0.78 0.604 0.40   1
Q19 0.75 0.569 0.43   1
Q20 0.68 0.466 0.53   1
Q21 0.68 0.467 0.53   1
Q22 0.70 0.488 0.51   1
Q23 0.64 0.406 0.59   1
Q24 0.66 0.440 0.56   1
Q25 0.76 0.583 0.42   1
Q26 0.66 0.430 0.57   1
Q27 0.47 0.222 0.78   1
Q28 0.73 0.529 0.47   1
Q29 0.69 0.474 0.53   1
Q30 0.70 0.496 0.50   1
Q31 0.65 0.417 0.58   1
Q32 0.73 0.539 0.46   1
Q34 0.65 0.428 0.57   1
Q35 0.74 0.542 0.46   1
Q36 0.77 0.590 0.41   1
Q37 0.76 0.572 0.43   1
Q38 0.80 0.647 0.35   1
Q39 0.72 0.515 0.48   1
Q40 0.75 0.564 0.44   1
Q41 0.30 0.089 0.91   1
Q42 0.77 0.594 0.41   1
Q43 0.74 0.542 0.46   1
Q44 0.80 0.642 0.36   1
Q45 0.72 0.516 0.48   1
Q46 0.15 0.021 0.98   1
Q47 0.52 0.269 0.73   1
Q49 0.72 0.520 0.48   1
Q50 0.53 0.281 0.72   1
```

```
Q51 0.62 0.380 0.62   1
Q52 0.76 0.577 0.42   1
Q53 0.73 0.535 0.46   1
Q54 0.75 0.568 0.43   1
Q55 0.27 0.075 0.93   1
Q56 0.82 0.670 0.33   1
Q57 0.72 0.521 0.48   1
Q58 0.73 0.533 0.47   1
Q59 0.76 0.579 0.42   1
Q60 0.47 0.224 0.78   1
Q61 0.72 0.511 0.49   1
Q62 0.65 0.424 0.58   1
Q63 0.77 0.590 0.41   1
Q64 0.71 0.503 0.50   1
Q65 0.61 0.373 0.63   1
Q66 0.71 0.498 0.50   1
Q68 0.68 0.459 0.54   1
Q69 0.57 0.323 0.68   1
Q70 0.56 0.309 0.69   1
Q71 0.74 0.544 0.46   1
Q72 0.54 0.294 0.71   1
Q73 0.64 0.405 0.59   1
Q74 0.60 0.359 0.64   1
Q76 0.74 0.544 0.46   1
Q77 0.77 0.598 0.40   1
Q78 0.66 0.442 0.56   1
Q79 0.60 0.365 0.64   1
Q80 0.68 0.466 0.53   1
Q81 0.65 0.419 0.58   1
Q82 0.61 0.368 0.63   1

                 PC1
SS loadings    29.96
Proportion Var  0.45

Mean item complexity =  1
Test of the hypothesis that 1 component is sufficient.

The root mean square of the residuals (RMSR) is  0.05
 with the empirical chi square  38864.3  with prob <  0

Fit based upon off diagonal values = 0.99
```

```
# PRINCIPAL AXIS FACTORING library(psych) fit <-
# factor.pa(items, nfactors=3, rotation='varimax') fit #
# print results
```

```
# PRINCIPAL COMPONENTS ANALYSIS entering raw data and
# extracting PCs from the correlation matrix fit <-
# princomp(items, cor=TRUE) summary(fit) # print variance
# accounted for loadings(fit) # pc loadings
# plot(fit,type='lines') # scree plot fit$scores # the
# principal components biplot(fit)
```

```
# DETERMINE NUMBER OF FACTORS TO EXTRACT library(nFactors)
# ev <- eigen(cor(items)) # get eigenvalues ap <-
# parallel(subject=nrow(items),var=ncol(items),
# rep=100,cent=.05) nS <- nScree(x=ev$values,
# aparallel=ap$eigen$qevpea) plotnScree(nS)
```

```
# MAXIMUM LIKELIHOOD FACTOR ANALYSIS entering raw data and
# extracting 3 factors, with varimax rotation fit <-
# factanal(items, 3, rotation='varimax') print(fit,
# digits=2, cutoff=.3, sort=TRUE) # plot factor 1 by factor
# 2 load <- fit$loadings[,1:2] plot(load,type='n') # set up
# plot text(load,labels=names(items),cex=.7) # add variable
# names
```

8.3 Confirmatory Factor Analysis (CFA)

The purpose of confirmatory factor analysis is, as the name implies, to confirm that items measure the factor(s) or trait(s) for which the items were designed and intended to measure.

In our use of CFA we used several statistics and fit indices:

- Root Mean Square Error of Approximation (RMSEA)
- Comparative Fit Index (CFI)
- Tucker–Lewis Index (TLI), and
- Standardized Root-Mean-Square Residual (SRMR)

Based on published criteria (Hu & Bentler, 1999; Wang & Wang, 2019), we used the following standards for good fit:

- CFI > 0.95
- TLI > 0.95
- RMSEA < 0.05
- SRMR < 0.08

We use the MLM estimator [Brown, 2006].

```
# ltm::factor.scores(items) # Computation of factor scores
# for grm, ltm, rasch and tpm models
```

```
# library(lavaan) MODEL WITH ONE COMMON FACTOR

# wb_1factor <- ' #start of model

# latent variable definitions (common factors) wellbeing =~
# NA*sense.peace + NA*sense.wonder + NA*sense.gratitude +
# NA*sense.purpose

# latent variable variances wellbeing ~~ 1*wellbeing

# latent variable covariances latent variable means

# manifest variable variances (uniquenesses) sense.peace ~~
# sense.peace sense.wonder ~~ sense.wonder sense.gratitude
# ~~ sense.gratitude sense.purpose ~~ sense.purpose

# manifest variable covariances (uniquenesses)

# manifest variable means sense.peace ~ 1 sense.wonder ~ 1
# sense.gratitude ~ 1 sense.purpose ~ 1 end of model

# CFA1 <- lavaan(wb_1factor, data=items, estimator = 'MLM')
# # estimate model summary(CFA1, standardized=TRUE,
# fit.measures=TRUE)
```

The results of CFA, using MLR estimation, confirmed the second order factor model, in line with the original structure of the scale (see Olufadi, 2017), because the values of the indices were above the acceptable threshold [2 (167) = 284.032, p < 0.001; RMSEA = 0.030 (90% CI = 0.024-0.036), CFI = 0.983, TLI = 0.981, SRMR = 0.023], compared to the unidimensional model [2 (170) = 316.949, p < 0.001; RMSEA = 0.034 (90% CI = 0.028-0.039), CFI = 0.978, TLI = 0.976, SRMR = 0.023] and a 3-uncorrelated-factors model [2 (170) = 2427.628, p < 0.001; RMSEA = 0.132 (90% CI = 0.127-0.136), CFI = 0.669, TLI = 0.631, SRMR = 0.345]. Based on RMSEA, CFI, TLI, and SRMR, the results indicated that the model provided satisfactory representations of the underlying structure of the psychological well-being construct. All items loaded significantly (ranging from 0.535 to 0.817) in relation to each first order factor, at a p < 0.01 significance level. These results of the CFA offer evidence that the four items measure one dimension of well-being.

8.4 Summary

Chapter 9

Scale Analysis

In this chapter, we examine the technical quality of the scales derived from the SHAS piloted items. We begin by defining scaling and discussing the importance of scale analysis. We report the results of our scale analysis from within the item response theory (IRT) framework. We conclude by considering what this information contributes to improved SHAS instrumentation.

9.1 The Item-Person Map

The item-person map (also known as the Wright map) is a tool for visualizing the results of a calibration of items and persons. The Wright map was originally developed for use in test development but it also helps illuminate the results of surveys [Bulut, 2021]. As Bulut [2021] explains:

"*The Partial Credit Model (PCM), which is an extension of the Rasch model for polytomous item responses, can be used for survey development and validation (Green & Frantom, 2002). We can analyze survey items using PCM and obtain item thresholds for each item. These thresholds indicate the magnitude of the latent trait required to select a particular response option (e.g., selecting disagree over strongly disagree). Using an item-person map, we can see thresholds for the items as well as the distribution of the latent trait for the respondents. This plot is quite useful for analyzing the alignment (or match) between the items and respondents answering the items.*"

```
# ITEM-PERSON MAP #1: FOR ALL THE SHAS ITEMS
source("code/itempersonmap.R")
library("mirt")
items <- read.csv("data/items.csv")
mod <- mirt(items, 1, itemtype = "Rasch", technical = list(removeEmptyRows = TRUE),
    verbose = FALSE)  # estimate the Partial Credit Model
itempersonmap(mod)  # create the item-person map
```

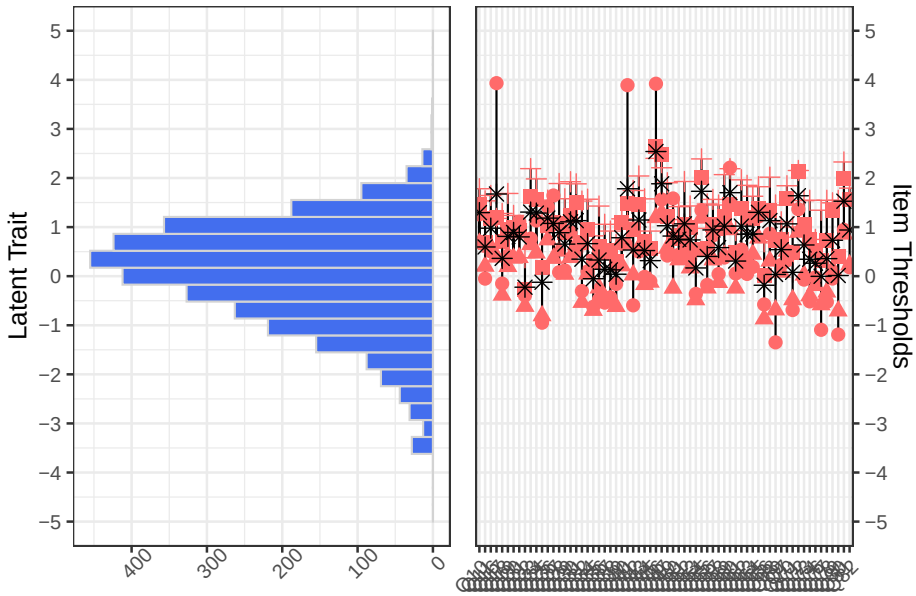


Figure 9.1: Item-Person Map: All Likert Items

Figure 9.1 is an item-person map of all 66 Likert SHAS items against all of the examinees. In the plot, red points indicate item thresholds for each item (three thresholds separating four response categories) and the asterisk indicates the average value of the thresholds for each item. The higher these thresholds, the more spiritual abuse the examinee has experienced. On the left-hand side of the plot, we also see the distribution of examinees along the latent trait (i.e., the construct of spiritual abuse).

This plot offers several insights at a general level. First, the plot of examinees on the left suggests that the latent trait of spiritual abuse is normally distributed, with a slight negative skew. Very few examinees reported very low or very high levels of exposure to spiritual abuse; most reported a moderate to slightly higher than average level of spiritual abuse. Second, with this plot it is difficult to distinguish individual items but it does show that they tend to clusteraround this "slightly higher than average" level of spiritual abuse, with none capturing a very low level of abuse and just a handful of items measuring much higher than average levels of spiritual abuse. Three items had response categories that captured very high levels of spiritual abuse.

9.2 Information curves

These provide important information about where along the scale of the latent trait the item or set of items provides the most reliable information. If there are too few "easy" or "difficult" items, then these extreme scores for examinees are less reliable.

9.3 Person Separation Reliability

9.4 Item Separation

9.5 Summary

Chapter 10

Discussion

References

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